

Partisan Tree Ascent: Edge-Labelled Games and Their Combinatorial Game Values

A. Researcher[†] B. Co-author[†] [†] *Department of Mathematics*

We define Partisan Tree Ascent, a two-player perfect-information game played on finite rooted binary trees whose edges carry labels L or R . Left (resp. Right) moves the shared token along any adjacent L -edge (resp. R -edge), always away from the root; the player unable to move loses (normal play convention). We compute the CGT values of representative trees under four edge-labelling regimes, prove a monotonicity theorem for pure-colour paths, and derive a temperature formula that exactly determines first-move urgency in compound positions.

1. Game definition

Definition 1. A *Partisan Tree Ascent position* is a pair (T, v) where T is a finite rooted binary tree with each edge labelled L or R , and $v \in V(T)$ is the current token location. **Left** moves v to any child reachable via an L -edge; **Right** moves via any R -edge. The player without a legal move **loses**.

The recursive CGT value at node v is

$$\mathcal{G}(v) = \{\mathcal{G}(c) \mid c \in \mathcal{L}(v) \mid \mathcal{G}(c) \mid c \in \mathcal{R}(v)\},$$

with $\mathcal{G}(\ell) = \{\mid\} = 0$ at every leaf ℓ .

2. Values by edge-labelling type

Table 1 catalogues values for representative paths and branching nodes.

Table 1: CGT values of selected positions. Temperature $t = -\infty$ denotes a number (cold game).

Position	Type	G	$t(G)$
L^1	I	1	$-\infty$
R^1	I	-1	$-\infty$
L^2	I	2	$-\infty$
LR	II	0	$-\infty$
LRL	II	1	$-\infty$
$\{0 \mid 0\}$	III	$*$	0
$\{0 \mid 1\}$	III	$\frac{1}{2}$	$-\infty$
$\{1 \mid -1\}$	III	± 1	1
$\{2 \mid -2\}$	III	± 2	2
$\{0 \mid *\}$	IV	$\uparrow$	0
$\{*\mid 0\}$	IV	$\downarrow$	0
$\{\uparrow \mid *\}$	IV	$\uparrow$	0

3. A monotonicity theorem

Theorem 2 (Monochromatic path values). *Let P_L^n denote the path $v_0 \xrightarrow{L} v_1 \xrightarrow{L} \dots \xrightarrow{L} v_n$ rooted at v_0 . Then $\mathcal{G}(P_L^n) = n$ for all $n \geq 0$. Symmetrically, $\mathcal{G}(P_R^n) = -n$.*

Proof. By induction on n . Base: $\mathcal{G}(P_L^0) = \{\mid\} = 0$. Inductive step: Left’s unique option from v_0 leads to the tail P_L^{n-1} with value $n-1$ by hypothesis; Right has no options. Hence $\mathcal{G}(P_L^n) = \{n-1 \mid\}$. By the *simplicity rule*, $\{n-1 \mid\} = n$, since n is the simplest number strictly greater than $n-1$ with no Right upper bound. $\square$

Corollary 3. P_L^n is a Left win unconditionally for all $n \geq 1$, regardless of who moves first.

Proof. For $n \geq 1$, $\mathcal{G}(P_L^n) = n > 0$, so Left wins under optimal play. Since Right cannot move at all, the outcome class is $\mathcal{L}$ irrespective of turn order—a stronger claim than “first-player wins.” $\square$

Remark 4. Purely monochromatic positions contribute only a constant integer to $\mathcal{G}(G_1 + \dots + G_k)$ and offer no strategic choice. All game-theoretic content lies in branching and mixed-colour types.

4. Temperature and urgency

For a game $G = \{G^L \mid G^R\}$ with $G^L < G^R$, the *temperature* and *mean value* satisfy

$$t(G) = \frac{m(G^L) - m(G^R)}{2}, \quad m(G) = \frac{m(G^L) + m(G^R)}{2}.$$

In any compound position $\sum_i G_i$, optimal play moves first in the component with the *largest* temperature.

Proposition 5. *For the symmetric switch $\pm n = \{n \mid -n\}$, we have $t(\pm n) = n$ and $m(\pm n) = 0$. The temperature of a Type-III symmetric branch equals the depth of its branches.*

Strategic consequence. In the compound game $(\pm 2) + (\pm 1) + *$, optimal play demands moving in ± 2 first (hottest, $t = 2$), then ± 1 ($t = 1$), treating $*$ last ($t = 0$). Deviating—e.g., playing in $*$ first—grants the opponent a swing of $2t(\pm 2) = 4$ half-moves, which is typically game-deciding.

5. Incomparability and the infinitesimal zoo

Type-IV positions expose the most striking feature of CGT: the game order is a *partial* order, not a total one.

Theorem 6 (Incomparability of $\uparrow$ and $*$). $\uparrow \parallel *$, i.e. $\uparrow$ and $*$ are incomparable under the standard CGT ordering.

Sketch. Compute $\uparrow - * = \uparrow + * = \{0 \mid *\} + * = \{** \mid *\}$. If Left moves first she plays to $** = 0$, winning; if Right moves first he plays to $*$, leaving $* + * = 0$, winning. Thus $\uparrow + *$ is *fuzzy* (outcome class $\mathcal{N}$), so $\uparrow \neq *$. Since $\uparrow > 0$ yet $* \parallel 0$, neither $\uparrow > *$ nor $\uparrow < *$ can hold. $\square$

This means that in positions containing both $\uparrow$ and $*$ summands, temperature-based heuristics fail entirely: no real-valued urgency ranking resolves the play correctly, and the full surreal-game machinery is required.

6. Open problems

- (i) **Complexity.** Is determining the outcome class of a Partisan Tree Ascent position in $\mathbf{P}$ or $\mathbf{PSPACE}$ -hard?

- (ii) **Realizability.** Characterise which CGT values arise as $\mathcal{G}(v)$ for some edge-labelled binary tree.
- (iii) **Generalisation.** Replace binary trees with k -ary trees and k player-colour edge labels. Do temperature strategies remain optimal, or does the richer structure force new phenomena?